MATH1112.02: Secret

nobody

Problem 1

1. (4 points)Let V be an inner product space, and $\{v_1, v_2, v_3\}$ be an orthonormal set in V.

   Then the norm
   $$\|5v_1 - 2v_2 + 3v_3\| = _____$$

2. (4 points)In R^3, let
   $$\mathbf{v}_1 = (3,1,2), \quad \mathbf{v}_2 = (5,3,4), \quad \mathbf{v}_3 = (1,1,1), \quad \mathbf{v}_4 = (4,2,3).$$
   The dimension of the subspace spanned by the above vectors is
   $$\dim\left(\text{span}\left\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\right\}\right) = _____$$

3. (4 points)Let $A = \begin{bmatrix} -2 & 1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$. All the singular values of A are _____,

4. (4 points)The inner product on the vector space M_2 is given by $\{A, B\rangle = tr\left(B^T A\right)$. Let
   $$C = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}$$
   The distance between C and D is $d(C, D) =$ _____

5. (6 points)The linear transformation $T : P_2 \to P_3$ is defined by
   $$T(p(x)) = xp(2x+1) - 2xp(x)$$
   for any $p(x) \in P_2$. Then
   $$\text{rank}(T) = _____, \text{nullity}(T) = _____$$

Problem 2

Let $V = C(-\infty, +\infty)$, the vector space of all continuous functions on $(-\infty, +\infty)$. Define operators T_1 and T_2 on V by

$$T_1(f(x)) = e^x f(x), \quad T_2(f(x)) = f(5x - 1)$$

for any $f(x) \in \mathbf{V}$.

1. (6 points) Prove that T_1 and T_2 are both linear operators.
2. (6 points) Take $g(x) = e^{-x}$. Find $(T_1 \circ T_2)(g(x))$ and $(T_2 \circ T_1)(g(x))$.

Problem 3

Let $V = P_2$. Define $\langle \cdot, \cdot \rangle$ on V by for any $p(x), q(x) \in V$

$$\langle p(x), q(x) \rangle = p(1)q(1) + p'(1)q'(1) + p''(1)q''(1)$$

1. (7 points) Prove that $\langle \cdot, \cdot \rangle$ is an inner product on V.

2. (8 points) Let
$$p_1(x) = 1, \quad p_2(x) = x, \quad p_3(x) = x^2.$$
Apply the Gram-Schmidt process to transform the standard basis $\{p_1(x), p_2(x), p_3(x)\}$ into an orthonormal basis $\{h_1(x), h_2(x), h_3(x)\}$.

   (Do NOT change the order of $\{p_1(x), p_2(x), p_3(x)\}$ in the process.)

Problem 4

(12 points) Find A^{2022}, where

$$A = \begin{bmatrix} \frac{5}{4} & \frac{3\sqrt{3}}{4} \\ \frac{3\sqrt{3}}{4} & -\frac{1}{4} \end{bmatrix}$$

Problem 5

(12 points) Suppose that $A \in M_3$ is symmetric and $\text{rank}(A) = 2$. Suppose that

$$A \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & 2 \\ 2 & 1 \end{bmatrix}$$

Find all the eigenvalues of A. For each eigenvalue, find an eigenvector corresponding to (Please give your explanation.)

Problem 6

Let V be a vector space of dimension n, and T be a linear operator on V. Suppose that there is a vector $v \in V$ such that $T^{n-1}(v) \neq 0$ and $T^n(v) = 0$.

1. (8 points) Prove that $B = \left\{\mathbf{v}, T(v), T^2(v), \ldots, T^{n-1}(v)\right\}$ is a basis for V.
2. (4 points) Find the matrix $[T]_B$ for T relative to the basis B.

Problem 7

7. Let $A \in M_n$. Suppose that ρ is the largest eigenvalue of $A^T A$.

   1. (6 points) Prove that $\|A\mathbf{x}\| \leq \sqrt{\rho}\|\mathbf{x}\|$ for any x $\in \mathbb{R}^n$, here $\|\cdot\|$ is the ordinary Euclidean norm on $\mathbb{R}^n$.

   2. (4 points) Prove that if $\rho < 1$, then $I - A$ is invertible.

   3. (5 points) Suppose that A is invertible. Prove that the matrix A can be written as $A = RH$, where R, H are matrices in M_n satisfying

      (1) R is orthogonal;

      (2) $H \in M_n$ is symmetric, and all its eigenvalues are positive,